

HEALTHCONNECT EXPERIENCE LAB

Final Project Summary Report

Track: Data Analytics

Week 8

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Project Summary

This summary captures the Week 8 Data Analytics contribution to the HealthConnect Clinic Experience Lab, focusing on final analytics output, validated findings, cross-track contribution, decision-support value and remaining limitations.

What Was Planned

- Review and carry forward validated Week 7 analytics findings.
- Confirm final KPIs and important dashboard figures.
- Finalize the analytics dashboard/report and decision-support content.
- Translate validated findings into business insights and actionable recommendations.
- Document the Data Analytics contribution to the wider HealthConnect solution.
- Complete relevant cross-track integration and prepare the analytics contribution for the final presentation.

What Was Completed

- Consolidated final KPIs: 48.5% no-show rate, 5.3% cancellation rate, 72.7% reminder coverage and 24.19-minute average waiting time.
- Carried forward the validated previous no-show finding.
- Documented key observed patterns involving previous no-show history, distance, reminder status and appointment type.
- Prepared business insights and evidence-informed recommendations.
- Prepared the Data Analytics contribution for final presentation.
- Documented the connection between Data Analytics, Project Management and Data Science.

What Was Validated

Item	Validation Outcome
Overall no-show rate	48.5%; consistent with the established analytics baseline.
Previous no-show group: 0	43.5% observed no-show rate.
Previous no-show group: 1–2	54.8% observed no-show rate.
Previous no-show group: 3+	68.8% observed no-show rate; interpret cautiously because the group is small.
Cross-track DA → PM validation	PASS; expected and actual grouped rates matched.

Most Important Finding

The most important validated analytical finding was the increasing observed no-show rate across previous no-show history groups: 43.5% for patients with no previous no-shows, 54.8% for those with one to two previous no-shows and 68.8% for those with three or more previous no-shows.

This suggests previous attendance behaviour is an important segment to consider when designing and testing attendance-support strategies. It remains an association rather than proof of causation.

Business Insights and Recommendations

- Consider targeted attendance support for patients with previous no-show history.
- Investigate accessibility and scheduling barriers for patients in the 21+ km distance group.
- Test broader reminder coverage and segment-specific reminder strategies.
- Monitor follow-up appointments as a higher-risk appointment segment.
- Review booking lead-time data quality before relying on it for downstream predictive modelling.

Cross-Track Collaboration

Data Analytics → Project Management

- Dependency: validated no-show evidence needed to support project testing and readiness tracking.
- Output provided: no-show analysis, KPI evidence and validation results.
- Output received: project testing context, evidence requirements and coordination structure.
- Integration activity: validation of the No-Show Rate by Previous No-Show Group.
- Result: expected and actual values matched; the tested KPI was recorded as PASS.

Data Analytics → Data Science

- Output provided: analytical patterns involving lead time, previous no-show history and appointment type × reminder channel.
- Output received: potential predictive features including high_lead_time_flag, chronic_no_show_flag and an appointment type × reminder channel interaction.
- Integration value: analytical observations were translated into potential inputs for predictive modelling.
- Limitation: Data Science figures were kept separate from the validated Week 7 analytics figures until differences in grouping or preprocessing are reconciled.

What Changed

- The analytics work moved from a dashboard-focused output into a consolidated decision-support package.
- Validated findings were separated from unvalidated or differently processed analytical outputs.
- The previous-no-show finding gained documented cross-track validation evidence.
- The analytics contribution was framed as an evidence layer for the wider HealthConnect solution.

Contribution to the Final HealthConnect Solution

The Data Analytics track contributes the evidence layer of the HealthConnect solution. It explains what is happening in appointment attendance, identifies important observed patterns and provides validated measures that other workstreams can use when considering predictive or operational actions.

- Data Analytics identifies patterns and measures appointment performance.
- Data Science can use relevant analytical patterns to inform potential predictive features.
- ML Engineering supports the technical workflow needed to operationalize technical components.
- Generative AI can support patient or administrative interaction within the broader solution.
- Project Management coordinates dependencies, evidence, risks and final delivery.

Testing and Refinement Outcome

Week 7 testing confirmed the main cross-track analytics KPI without requiring a numerical correction. The main refinement was improved documentation of validation evidence and clearer separation between verified results and work requiring further evidence.

Limitations and Remaining Issues

- The analytics findings are observational and do not prove causation.
- The 3+ previous no-show group contains relatively few appointments.
- Some dataset variables contain missing values.
- Booking lead-time data requires further quality review before being treated as a validated predictive feature.
- Available evidence does not independently establish complete end-to-end validation across every technical workstream.
- Recommendations should be tested operationally before effectiveness is assumed.

Final Readiness Position

The Data Analytics contribution has been consolidated and the principal tested KPI has been validated. The analytics component is suitable for inclusion in the final HealthConnect presentation and decision-support package. Project-level readiness should continue to be based on evidence from all required workstreams rather than inferred from the Data Analytics validation alone.

Final Presentation Focus

- Introduce the HealthConnect problem and the Data Analytics role.
- Explain the final KPIs and what they reveal about appointment performance.
- Present the validated previous-no-show finding and other important observed patterns.
- Explain testing, validation and refinement rather than simply showing the dashboard.
- Show how Data Analytics connects with Project Management and Data Science.
- Explain the business value and limitations of the analytical contribution.
- Close with how the analytics evidence supports the wider HealthConnect solution.

Key Takeaway

Useful analytics goes beyond creating charts. The process involves understanding the data, validating the numbers, identifying meaningful patterns, acknowledging limitations and translating evidence into decisions that can support the wider solution.